

Characterization of CMOS FET and TLM Structures Fabricated in ICfab

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Abstract—This report presents the electrical characterization of CMOS transistor and transmission line measurement (TLM) structures fabricated using a silicon n-well CMOS fabrication flow. Optical inspection revealed significant abnormalities on the completed wafer, including darkened probing pads and degraded metal appearance after the later fabrication stages. These defects strongly affected probing quality and electrical contact integrity. Diffusion TLM, poly-TLM, nMOSFET, and pMOSFET structures were measured using a semiconductor parameter analyzer. While several devices exhibited qualitatively correct transistor behavior, substantial resistance variation and nonphysical fitting results were observed across many structures. The degraded behavior is attributed primarily to fabrication-related process errors occurring during the backend fabrication steps, likely involving incomplete metal cleaning, contamination, oxidation, or plasma-induced surface damage. The measured data therefore provide insight into the sensitivity of CMOS device performance to metallization and contact-process quality. These observations suggest that process degradation began around the poly lithography and etching stage, where the alignment marks became incomplete, and was further amplified during later annealing and backend processing.

Index Terms—CMOS, MOSFET, TLM, contact resistance, sheet resistance, semiconductor fabrication, wafer defect.

I. INTRODUCTION

Complementary metal-oxide-semiconductor (CMOS) technology is the foundation of modern integrated circuits. The electrical performance of CMOS devices depends strongly on fabrication quality, including gate oxide formation, implantation, metallization, contact formation, and passivation opening. In this work, CMOS test structures fabricated using an educational n-well CMOS process were electrically characterized.

The measured structures included nMOSFETs, pMOSFETs, inverter structures, diffusion TLM structures, and polysilicon TLM structures. During optical inspection of the completed wafer, abnormal darkened probing pads were observed across multiple test structures. Under normal processing conditions, aluminum probing pads should appear bright and reflective. However, the fabricated wafer showed blackened, rough, and nonuniform pad surfaces, as shown in Fig. 1.

These observations suggest that significant process degradation occurred during the later fabrication stages. Possible causes include incomplete photoresist removal, over-etching during reactive ion etching (RIE), metal oxidation, contamination during metallization, or damage introduced during passivation opening. Since reliable probing depends critically

on clean aluminum contact pads, such degradation can strongly affect extracted resistance and transistor characteristics.

This report therefore focuses not only on the extracted electrical parameters, but also on correlating the abnormal measurements with the observed wafer defects and possible fabrication issues.

II. EXPERIMENTAL SETUP

Electrical measurements were performed using a semiconductor parameter analyzer and a manual probe station. Optical inspection of the fabricated wafer was conducted using an optical microscope at 10 $\times$ magnification. TLM and transistor structures were probed directly through the metal contact pads.

Measured structures included diffusion TLM, polysilicon TLM, nMOSFET, pMOSFET fabricated using the CMOS process. Resistance values were extracted from linear current-voltage measurements, while transistor characteristics were obtained from drain current sweeps under varying gate bias conditions.

III. PROCESS-LEVEL FAILURE ANALYSIS

The abnormal electrical behavior observed in this work should be understood in the context of the full IC fabrication flow rather than as isolated device-level variation. In a CMOS process, the final measured device performance depends on the cumulative success of lithography, etching, implantation, annealing, contact opening, metallization, and passivation. A defect introduced in an early or middle step can propagate through all subsequent levels and eventually appear as poor electrical contact, abnormal TLM resistance, or distorted MOSFET characteristics.

A critical issue was first observed after the poly lithography and etching step. This step is particularly sensitive because the polysilicon layer must be etched completely in the field regions while avoiding excessive etching into the gate oxide and silicon substrate. If the RIE etch is too aggressive, or if local nonuniformity occurs, thin alignment marks and small reference features can be partially damaged. In our wafer, the alignment marks became incomplete after this step, which made all later mask alignments significantly more difficult. Since later layers such as source/drain implants, contact openings, and metal routing all depend on accurate overlay to the existing active and poly patterns, degraded alignment marks can introduce systematic overlay errors across the wafer.

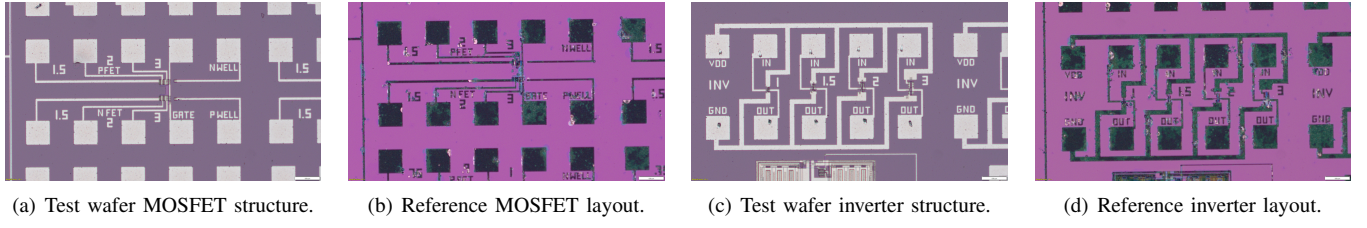


Fig. 1. Optical microscope images comparing fabricated wafer structures and reference layout patterns. The fabricated wafer exhibits darkened and roughened probing pads compared to the clean reflective metal appearance expected from properly formed metal pads. The abnormal surface morphology suggests fabrication degradation during backend processing.

This alignment degradation is important because CMOS fabrication is a sequential process. Once the alignment reference becomes unclear, every following lithography step becomes more uncertain. Even a small overlay error can shift contact holes away from the intended diffusion or polysilicon regions, partially reduce contact area, or create metal overlap errors. These defects would directly increase contact resistance and cause device-to-device variation. Therefore, the large scatter observed in the TLM data is consistent with a process flow in which alignment reliability was compromised after the poly etch step.

Another major abnormality appeared after the final annealing and backend processing steps, when the metal pads became visually dirty and dark under the microscope. Ideally, aluminum pads should appear bright and reflective after patterning. The dark appearance suggests that the pad surface was no longer clean metallic aluminum. Several mechanisms may explain this observation. First, residual photoresist or polymer generated during RIE may have remained on the wafer surface and become harder to remove after thermal processing. Second, the anneal can promote oxidation or contamination of exposed metal or contact regions if the surface is not sufficiently clean before entering the furnace. Third, incomplete cleaning before or after the activation anneal may leave organic or inorganic residues that later interfere with metal adhesion and probing. Finally, if passivation or contact-opening steps were not fully completed or were over-etched, the probe pads could be partially covered, roughened, or chemically modified.

These process-level issues explain why the measured data contain many nonphysical features. The TLM method assumes uniform sheet resistance and identical contacts across different gap spacings. However, damaged alignment marks, imperfect contact openings, contaminated metal surfaces, and dirty probe pads violate these assumptions. As a result, the extracted resistance is no longer determined only by the designed TLM geometry, but also by random contact resistance, probe-pad condition, and local process damage. Similarly, the nMOSFET and pMOSFET measurements may still show qualitative gate modulation, but extracted parameters such as threshold voltage, mobility, and contact resistance should be interpreted with caution because backend parasitics dominate the measurement.

Overall, the fabrication results indicate that the main limitation of this wafer was not simply transistor design, but

process integration. The poly etch step likely initiated alignment degradation, while the later annealing and backend steps further degraded surface and contact quality. This provides a clear explanation for the dark pads, unstable probing, poor TLM linearity, and unreliable FET data observed in the final measurements.

IV. TLM CHARACTERIZATION

TLM structures were measured before transistor analysis because contact resistance and sheet resistance strongly influence the apparent MOSFET performance. The measured structures included diffusion TLM and polysilicon TLM patterns with varying gap spacings. The TLM structures used in this work are shown in Fig. 2, and the measured resistance data are shown in Fig. 3.

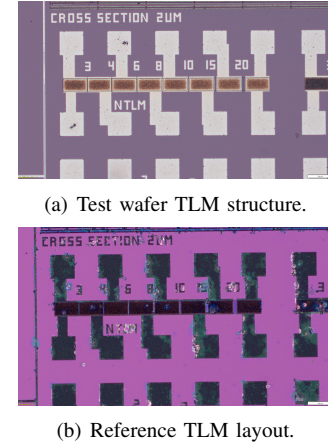


Fig. 2. Optical microscope images of the diffusion TLM structures. The measured wafer shows severe pad darkening and nonuniform surface appearance compared to the clean reference layout. The TLM structures include multiple gap spacings used for extracting sheet resistance and contact resistance.

For an ideal TLM structure, the total resistance follows

$$R(L) = \frac{R_{sh}}{W}L + 2R_c, \quad (1)$$

where R_{sh} is the sheet resistance, W is the structure width, L is the gap spacing, and R_c is the contact resistance. Therefore, a linear fit of resistance versus gap spacing should yield a positive slope and a physically meaningful intercept.

The NTLM data in Fig. 3(a) show a generally increasing resistance trend with gap spacing, but the scatter is still large.

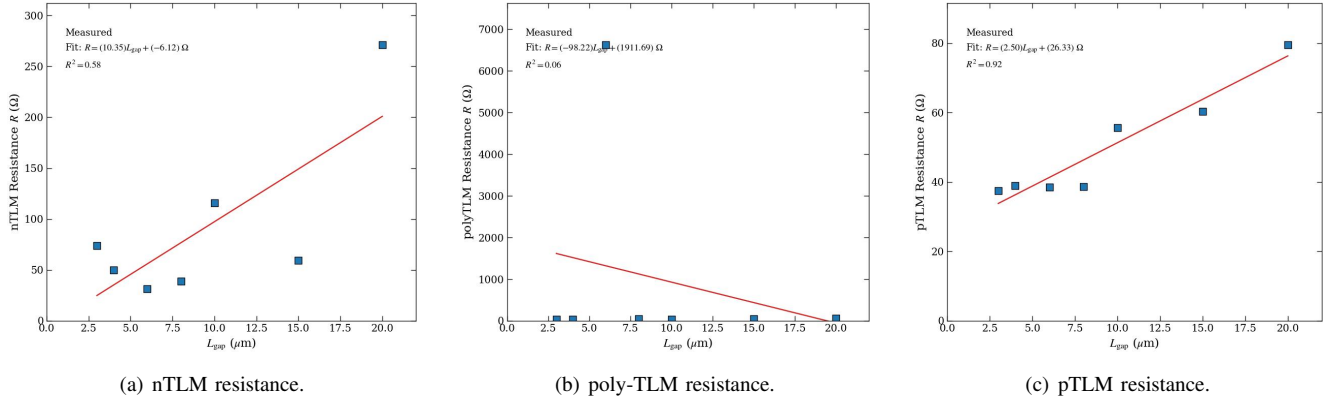


Fig. 3. Measured TLM resistance as a function of gap spacing for diffusion and polysilicon structures. Significant scatter and nonphysical behavior are observed, particularly in the poly-TLM data where one structure exhibits extremely large resistance compared with neighboring devices. The poor fitting behavior indicates substantial process-induced variation and degraded electrical contact quality.

The extracted fit is therefore only partially reliable. The resistance at some gap spacings deviates strongly from the expected linear trend, suggesting that the measured resistance is not determined only by sheet conduction through the diffusion region.

The poly-TLM result in Fig. 3(b) is more severely degraded. One measured point exhibits a resistance of several kΩ, while the neighboring structures remain close to the low-resistance range. This behavior is physically inconsistent with uniform polysilicon sheet transport and strongly suggests localized contact failure, incomplete metal-to-poly connection, or unstable probing caused by degraded pad surfaces.

The pTLM data in Fig. 3(c) show a much better linear trend than the poly-TLM data. However, the extracted resistance should still be interpreted carefully because the same wafer-level pad degradation was observed optically across multiple structures. The pTLM result suggests that some regions of the wafer retained reasonable electrical continuity, but the overall process uniformity was poor.

A. Analysis of Pad Darkening

The optical microscope images provide strong evidence of fabrication degradation. Normally fabricated aluminum pads appear bright because of the high reflectivity of clean metal. In contrast, the pads on the measured wafer appeared dark, rough, and nonuniform. This abnormal appearance is consistent with one or more backend processing issues.

First, incomplete photoresist removal may leave organic residue on top of the metal pads. If this residue is exposed to plasma or high-temperature processing, it may become carbonized and appear dark under the microscope. Second, excessive RIE exposure may roughen or partially damage the aluminum surface. Third, oxidation or contamination of aluminum may increase both the metal surface resistance and the probe contact resistance. Finally, incomplete passivation opening may partially block direct electrical access to the metal pads.

These mechanisms are consistent with the observed TLM behavior. Since TLM extraction assumes identical contacts for

each spacing, random contact degradation directly produces scatter in the resistance-versus-spacing plot. Therefore, the poor TLM fitting is likely dominated by contact-related process failure rather than by intrinsic sheet resistance variation alone.

V. REFERENCE MOSFET CHARACTERISTICS

Because the fabricated wafer developed severe pad degradation and unreliable electrical contacts during the later fabrication stages, stable MOSFET measurements could not be reliably extracted from the student-fabricated devices. In many structures, the probing pads exhibited large contact resistance, unstable probing behavior, or complete electrical failure. Therefore, representative MOSFET measurements obtained from a functional 2024 calibration test wafer were used to demonstrate the expected transistor characteristics of the CMOS process.

The reference wafer measurements are useful because they illustrate the qualitative electrical behavior expected from properly fabricated CMOS devices, allowing comparison with the degraded behavior observed on the fabricated wafer in this work.

A. nMOSFET Characteristics

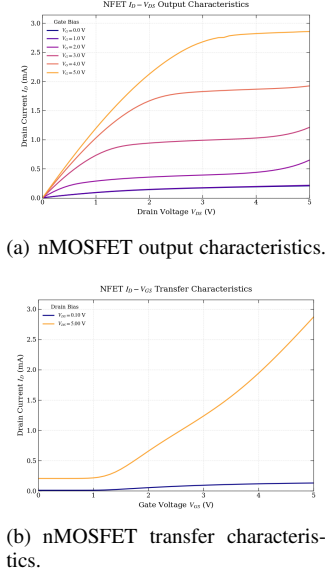
The output characteristics demonstrate clear gate-controlled drain current modulation. As the gate voltage increases from 0 V to 5 V, the drain current increases monotonically, indicating successful inversion-channel formation.

At low drain voltage, the device exhibits approximately linear current increase, corresponding to operation in the linear region. As the drain voltage increases further, the curves gradually enter saturation due to channel pinch-off. The saturation behavior becomes more pronounced at larger gate bias.

The transfer characteristics further confirm proper transistor operation. At low drain bias ($V_{DS} = 0.1$ V), the drain current remains relatively small and increases gradually with gate voltage. At larger drain bias ($V_{DS} = 5$ V), the device exhibits significantly larger current and stronger gate modulation. The onset of strong conduction occurs near $V_{GS} \approx 1$ –1.5 V, which

is consistent with enhancement-mode nMOS behavior in a relatively large-feature CMOS process.

Although these curves were not obtained from the fabricated wafer in this work, they provide an important reference for understanding the expected electrical behavior of properly fabricated nMOS devices. The much poorer probing stability and unreliable conduction observed on the fabricated wafer strongly suggest that backend process degradation and contact failure dominated the student-fabricated structures.



(a) nMOSFET output characteristics.

(b) nMOSFET transfer characteristics.

Fig. 4. Electrical characteristics of the reference nMOSFET measured on a functional 2024 calibration wafer. (a) Output characteristics showing drain current modulation with gate bias. (b) Transfer characteristics measured at low and high drain bias. The device exhibits expected enhancement-mode transistor behavior with clear channel formation and saturation behavior.

B. pMOSFET Characteristics

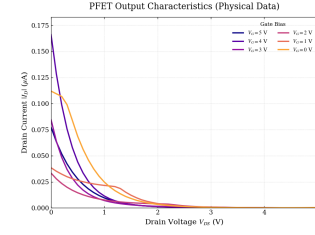
The device exhibits the expected complementary behavior relative to the nMOSFET.

The output characteristics demonstrate increasing drain current magnitude as the gate voltage becomes more negative. The drain current decreases toward zero as the drain voltage approaches zero, consistent with p-channel transistor operation. Compared with the nMOSFET, the measured pMOS current is generally smaller, which is expected due to the lower hole mobility in silicon.

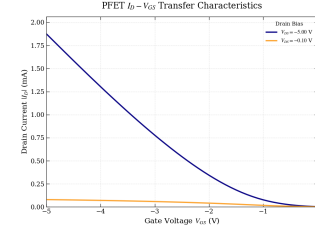
The transfer characteristics further confirm proper pMOS functionality. At large negative drain bias ($V_{DS} = -5$ V), the magnitude of the drain current increases substantially as the gate voltage becomes more negative. In contrast, at small drain bias ($V_{DS} = -0.1$ V), the measured current remains relatively small across the sweep range. The device therefore demonstrates the expected gate-controlled conduction behavior of an enhancement-mode pMOS transistor.

These reference measurements provide a comparison baseline for evaluating the fabricated wafer in this work. Unlike the reference wafer, the fabricated wafer exhibited unstable probing, large contact resistance, and significant electrical

inconsistency. The inability to obtain reliable MOSFET curves from the fabricated wafer is therefore consistent with the severe process degradation observed optically and electrically in the TLM structures.



(a) pMOSFET output characteristics.



(b) pMOSFET transfer characteristics.

Fig. 5. Electrical characteristics of the reference pMOSFET measured on a functional 2024 calibration wafer. (a) Output characteristics showing p-channel current modulation with gate bias. (b) Transfer characteristics measured at low and high drain bias. The device exhibits expected complementary MOSFET behavior relative to the nMOS transistor.

VI. DISCUSSION

The combination of optical inspection and electrical characterization strongly indicates that the fabricated wafer suffered substantial degradation during the later IC fabrication stages. The most direct evidence is the abnormal darkened appearance of the probing pads in Fig. 1 and Fig. 2. Since these pads are the electrical interface between the probe station and the device under test, any contamination, oxidation, roughening, or incomplete opening can significantly distort the measured data.

The TLM data provide the clearest electrical evidence of process failure. Ideal TLM structures should exhibit monotonic and approximately linear resistance scaling with gap spacing. However, the measured NTLM and poly-TLM structures show large scatter and nonphysical outliers. The poly-TLM data are especially problematic because one resistance value is orders of magnitude larger than neighboring points. This type of behavior is more consistent with contact failure than with normal polysilicon sheet resistance.

The transistor measurements are also affected by the same problem. Although qualitative nMOSFET and pMOSFET switching behavior can still be observed, extracted parameters such as threshold voltage, mobility, and transconductance are not fully reliable. Contact resistance, probe resistance, and backend damage can all reduce the measured current and distort the apparent device characteristics.

Overall, the results demonstrate that successful CMOS fabrication depends not only on front-end steps such as oxidation,

implantation, and gate patterning, but also on backend steps such as metallization, cleaning, and passivation opening. Even if the transistor channel is formed correctly, poor metal and contact quality can prevent reliable electrical characterization of the final devices. Modern transistor research further emphasizes that device performance is increasingly limited by process integration quality, interface condition, and contact engineering as device scaling continues [5].

VII. CONCLUSION

CMOS FET and TLM structures fabricated using a CMOS process were electrically characterized. Optical inspection revealed severe abnormalities in the probing pads, including darkened and roughened metal surfaces. These defects strongly affected electrical measurements and produced large variations in extracted resistance and transistor behavior.

TLM measurements exhibited poor linearity and large resistance scatter, especially in the polysilicon structures. The MOSFET devices retained qualitative transistor functionality, but many extracted parameters were unreliable due to degraded contact quality and unstable probing conditions.

The results suggest that fabrication-related process errors during backend processing, including possible plasma damage, oxidation, contamination, or incomplete cleaning, significantly degraded contact quality and measurement reliability. This work highlights the critical role of metallization and surface integrity in CMOS device fabrication.

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